A Barvinok–Pataki Theorem for Archimedean Order Unit Spaces

Timo Ziegler,[1] Tobias J. Osborne,[1] Martin Plavala,[2]
René Schwonnek,[1] Lennart Binkowski,[1] Lukas Hantzko,[2]
Henrik Wilming,[1] Alexander Stottmeister[3]

[1]AG Osborne, Institut für Theoretische Physik, Leibniz Universität Hannover
[2]AG Raussendorf, Institut für Theoretische Physik, Leibniz Universität Hannover
[3]Quantum Information Group, Institut für Theoretische Physik, Leibniz Universität Hannover

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Abstract

We prove a generalisation of the Barvinok–Pataki rank bound to Archimedean order unit spaces equipped with a pointed cone. The classical result bounds the rank of extreme points of spectrahedra (feasible sets of semidefinite programs); our generalisation bounds the face dimension of feasible points under affine constraints in an arbitrary pointed cone, including the infinite-dimensional case. The face-dimension bound $d_F \leq m$ uses only pointedness of the cone and rank–nullity; finite-dimensionality of d_F in infinite dimensions follows from a result of Weis–Shirokov that requires only a real vector space and a convex set. When specialised to C*-algebras via the Alfsen–Shultz face–commutant correspondence, this recovers the operator-algebraic Barvinok–Pataki bound $\sum n_j^2 \leq m + 1$.

1 Preliminaries

Definition 1 (Ordered vector space with cone)**.** Let V be a real vector space. A *cone* is a subset $V^+ \subseteq V$ that is closed under addition and non-negative scalar multiplication. The pair (V, V^+) is an *ordered vector space* with partial order $x \leq y$ iff $y - x \in V^+$.

Definition 2 (Pointed cone)**.** A cone V^+ is *pointed* (or *proper*) if $V^+ \cap (-V^+) = \{0\}$, i.e. the only element v with both $v \geq 0$ and $-v \geq 0$ is $v = 0$.

Definition 3 (Order unit and Archimedean property)**.** An element $e \in V^+$ is an *order unit* if for every $v \in V$ there exists $n \in \mathbb{N}$ such that $ne - v \in V^+$ (i.e. $v \leq ne$). The cone V^+ is *Archimedean* if, whenever $ne + v \in V^+$ for all $n \in \mathbb{N}$, it follows that $v \in V^+$. The triple (V, V^+, e) is called an *Archimedean order unit space* (AOU space).

Definition 4 (Face of a cone)**.** Let V^+ be a cone in V and $x \in V^+$. The *face of V^+ generated by x* is

$$\mathrm{face}_{V^+}(x) \;=\; \left\{ w \in V^+ \;\middle|\; \exists\, \lambda > 0 \colon x - \lambda w \in V^+ \right\}.$$

Equivalently, $\mathrm{face}_{V^+}(x)$ consists of all elements of V^+ that can be "majorised" by a positive multiple of x. This is the smallest face of V^+ containing x in its relative interior.

Remark 5. In a pointed cone, $\mathrm{face}_{V^+}(x) = \{0\}$ if and only if $x = 0$. For $x \neq 0$, the face is at least one-dimensional (it contains the ray $\{\lambda x : \lambda \geq 0\}$). For the PSD cone PSD_n, the face generated by a rank-r matrix X is isomorphic to PSD_r, with $\dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$ (real symmetric case) or r^2 (complex Hermitian case).

2 The Classical Barvinok–Pataki Bound

We first recall the classical proof for PSD matrices, as it motivates each step of the generalisation.

Theorem 6 (Barvinok–Pataki, PSD case)**.** *Let $A_1, \dots, A_m \in \mathrm{Sym}_n(\mathbb{R})$ and $\beta_1, \dots, \beta_m \in \mathbb{R}$. Consider the spectrahedron*

$$C = \{\, X \in \mathrm{PSD}_n \mid \langle A_j, X\rangle = \beta_j,\ j = 1, \dots, m \,\}.$$

If $X \in C$ is an extreme point of C, then $\frac{r(r+1)}{2} \le m$ where $r = \mathrm{rank}(X)$.

Proof. Write $X = VBV^\top$ with $V \in \mathbb{R}^{n\times r}$, $B \in \mathrm{PSD}_r$ strictly positive definite. Define the compressed constraints $\hat{A}_j = V^\top A_j V \in \mathrm{Sym}_r$ and the linear map

$$\varphi\colon \mathrm{Sym}_r \to \mathbb{R}^m, \qquad [\varphi(D)]_j = \langle \hat{A}_j, D\rangle.$$

By rank–nullity, $\dim(\ker\varphi) \ge \frac{r(r+1)}{2} - m$. If $\ker\varphi \ne \{0\}$, choose $D \ne 0$ in $\ker\varphi$. Then $B(t) = B + tD$ satisfies all constraints for every t.

Since $B \succ 0$ and D is a non-zero symmetric matrix with at least one negative eigenvalue (replacing D by $-D$ if necessary), the map $t \mapsto \lambda_{\min}(B + tD)$ is continuous, positive at $t = 0$, and decreasing for $t > 0$. Let $t_0 > 0$ be the first value where $\lambda_{\min}(B + t_0 D) = 0$. Then $B(t_0) \succeq 0$ with $\mathrm{rank}(B(t_0)) < r$, and $VB(t_0)V^\top \in C$.

This contradicts X being an extreme point of C (since X is a non-trivial convex combination of $VB(t_0)V^\top$ and $VB(-t_0')V^\top$ for suitably chosen t_0'), unless $\ker\varphi = \{0\}$, which forces $\frac{r(r+1)}{2} \le m$. □

3 Generalisation to Pointed Cones

The key observation is that the proof above uses only three ingredients:

1. **Rank–nullity** for the constraint map restricted to the face;
2. **Pointedness** of the cone, to guarantee that a line through an interior point of a face must exit the cone;
3. **Finite-dimensionality** of the face, to apply rank–nullity with finite dimensions.

Theorem 7 (Barvinok–Pataki for pointed cones)**.** *Let V be a real vector space, $V^+ \subseteq V$ a pointed cone, and $\alpha_1, \dots, \alpha_m \in V^*$ linear functionals. Define*

$$C = \{\, x \in V^+ \mid \alpha_j(x) = b_j,\ j = 1, \dots, m \,\}$$

for given $b_1, \dots, b_m \in \mathbb{R}$. Let $x \in C$ and let $F = \mathrm{face}_{V^+}(x)$ be the face of V^+ generated by x. Assume that $d_F := \dim(\mathrm{span}(F))$ is finite. Define the restricted constraint map

$$\varphi_F\colon \mathrm{span}(F) \to \mathbb{R}^m, \qquad [\varphi_F(v)]_j = \alpha_j(v).$$

If x is an extreme point of C, then

$$\boxed{d_F \le m.}$$

Equivalently, $\dim(F) \le m - 1$ (where $\dim(F)$ is the dimension of F as a convex set, i.e. $\dim(\mathrm{aff}(F))$).

Proof. By rank–nullity applied to φ_F,

$$d_F \;=\; \dim(\ker\varphi_F) + \dim(\operatorname{im}\varphi_F) \;\le\; \dim(\ker\varphi_F) + m.$$

It suffices to show that $\ker\varphi_F = \{0\}$ when x is extreme in C. Suppose for contradiction that there exists $D \in \ker\varphi_F$, $D \neq 0$.

Step 1: The perturbation preserves feasibility. Since $D \in \operatorname{span}(F) \cap \ker\varphi_F$, we have $\alpha_j(D) = 0$ for all j. Therefore, for any $t \in \mathbb{R}$:

$$\alpha_j(x + tD) = \alpha_j(x) + t\,\alpha_j(D) = b_j.$$

So $x + tD$ satisfies all affine constraints whenever it lies in V^+.

Step 2: Pointedness forces exit from the cone. Since V^+ is pointed, we cannot have both $D \in V^+$ and $-D \in V^+$ (this would imply $D = 0$). Therefore, at least one of the half-lines $\{x + tD : t > 0\}$ or $\{x - tD : t > 0\}$ must eventually leave V^+.

More precisely: since x lies in the relative interior of F, for sufficiently small $|t|$ we have $x + tD \in F \subseteq V^+$ (as $D \in \operatorname{span}(F)$ and x is an interior point of F). But the line cannot remain in V^+ in both directions indefinitely, since this would imply D and $-D$ are both in V^+ (by rescaling), contradicting pointedness.

Step 3: Walking to a smaller face. Let

$$t^+ = \sup\{t > 0 : x + tD \in V^+\}, \qquad t^- = \sup\{t > 0 : x - tD \in V^+\}.$$

By Step 2, at least one of t^+, t^- is finite. Say $t^+ < \infty$ (the other case is symmetric).

If V^+ is closed (which holds in finite dimensions, or under the Archimedean assumption in the norm topology), then $x^* := x + t^+D \in V^+$, and x^* lies on the boundary of V^+ relative to $\operatorname{span}(F)$. This means x^* belongs to a *proper face* $F' \subsetneq F$, so $\dim(F') < \dim(F)$. Moreover, x^* satisfies all constraints, so $x^* \in C$.

Step 4: Contradiction with extremality. We have $x^* = x + t^+D \in C$ and (if $t^- > 0$, which holds since x is interior to F) also $x^{**} = x - \varepsilon D \in C$ for small $\varepsilon > 0$. Then

$$x = \frac{\varepsilon}{t^+ + \varepsilon}\, x^* + \frac{t^+}{t^+ + \varepsilon}\, x^{**},$$

exhibiting x as a non-trivial convex combination of two distinct elements of C. If x were an extreme point of C, this is a contradiction.

Therefore $\ker\varphi_F = \{0\}$, and we conclude $d_F \le m$. □

4 Application to C*-Algebras and AOU Spaces

Corollary 8 (Barvinok–Pataki for AOU spaces)**.** *Let (V, V^+, e) be an Archimedean order unit space with $\dim(V) < \infty$. Let $\alpha_1, \ldots, \alpha_m \in V^*$ and consider the state-like set*

$$C = \big\{\, x \in V^+ \;\big|\; \alpha_j(x) = b_j,\ j = 1, \ldots, m \,\big\}.$$

For any extreme point x of C, the face of V^+ generated by x satisfies $\dim(\operatorname{face}_{V^+}(x)) \le m - 1$.

Proof. The Archimedean property implies the cone V^+ is closed (in the order-unit norm) and pointed. In finite dimensions, every face is finite-dimensional. Apply Theorem 7. □

Corollary 9 (Barvinok–Pataki for C*-algebras)**.** *Let A be a unital C^*-algebra and $a_1, \ldots, a_m \in A_{\mathrm{sa}}$. Consider the affinely constrained state space*

$$C = \big\{ \varphi \in S(A) \;\big|\; \varphi(a_j) = \beta_j,\ j = 1, \ldots, m \big\},$$

where $S(A) = \{\varphi \in A^* : \varphi \geq 0,\ \varphi(\mathbf{1}) = 1\}$. If φ is an extreme point of C, then

$$\dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} \;\leq\; m + 1,$$

where π_φ is the GNS representation of φ. Equivalently, if $\pi_\varphi(A)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ (Artin–Wedderburn), then $\sum_j n_j^2 \leq m + 1$.

Proof. The state space $S(A)$ is a base for the cone A_+^* (positive functionals), cut out by the normalisation constraint $\varphi(\mathbf{1}) = 1$. The total number of affine constraints on A_+^* is $m + 1$ (the m given constraints plus normalisation).

By the Alfsen–Shultz face–commutant correspondence, the face of $S(A)$ generated by φ is affinely isomorphic to the state space $S(\pi_\varphi(A)')$ of the commutant. In particular,

$$\dim\big(\mathrm{face}_{S(A)}(\varphi)\big) = \dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} - 1.$$

Applying Theorem 7 with the $m+1$ constraints on A_+^* gives $\dim(\mathrm{span}(\mathrm{face}(\varphi))) \leq m+1$, hence $\dim_{\mathbb{R}}(\pi_\varphi(A)')_{\mathrm{sa}} \leq m + 1$. □

5 The Infinite-Dimensional Case and the Role of d_F

Theorem 7 assumes that $d_F = \dim(\mathrm{span}(F))$ is finite. In infinite dimensions this is a non-trivial hypothesis: the face of a cone generated by a point can be infinite-dimensional. We now show that this hypothesis is *automatically satisfied* whenever the number of constraints is finite, using a result of Weis–Shirokov that is purely convex-geometric.

Theorem 10 (Weis–Shirokov [4, Theorem 3.3])**.** *Let V be a real vector space, $K \subseteq V$ a convex set, and $f\colon K \to \mathbb{R} \cup \{+\infty\}$ a generalised affine map. Let C denote a sublevel set $\{x \in K : f(x) \leq \alpha\}$ or a level set $\{x \in K : f(x) = \alpha\}$, and let $x \in C$. If the face $F_C(x)$ of C generated by x has dimension $m \in \mathbb{N}_0$, then the face $F_K(x)$ of K generated by x has dimension m or $m + 1$.*

The multi-constraint version follows by induction (Corollary 5 of [4]): if C is the intersection of ℓ sublevel and/or level sets of generalised affine maps, and $F_C(x)$ has dimension m, then $\dim(F_K(x)) \in \{m, m + 1, \ldots, m + \ell\}$.

Corollary 11 (Finiteness of d_F)**.** *Let K be a convex set in a real vector space V, and let C be defined by ℓ generalised affine constraints on K. If x is an extreme point of C, then*

$$\dim\big(F_K(x)\big) \;\leq\; \ell.$$

In particular, the face of K generated by any extreme point of C is finite-dimensional, regardless of $\dim(V)$.

Proof. Since x is extreme in C, the face $F_C(x) = \{x\}$ has dimension 0. By the multi-constraint version of Theorem 10, $\dim(F_K(x)) \in \{0, 1, \ldots, \ell\}$. □

Remark 12 (No operator-algebraic assumptions needed)**.** Theorem 10 and Corollary 11 require only a real vector space V and a convex subset $K \subseteq V$. In particular:

- No cone structure, pointedness, or Archimedean property is needed to establish that d_F is finite.

- No C*-algebraic or operator-algebraic notions are needed.
- The operator-algebraic machinery (GNS representation, face–commutant correspondence, Artin–Wedderburn decomposition) is used only in a second step: to *interpret* the finite-dimensional face algebraically as a commutant and extract the multiplicity bound $\sum n_j^2 \leq m+1$.

This clarifies the logical structure of the infinite-dimensional case: Weis–Shirokov provides the purely geometric bound on d_F, and the Alfsen–Shultz correspondence translates it into algebra.

Combining Corollary 11 with Theorem 7, we obtain the following infinite-dimensional version of the main theorem that does not require an a priori finiteness assumption on d_F.

Corollary 13 (Barvinok–Pataki for infinite-dimensional pointed cones)**.** *Let V be a (possibly infinite-dimensional) real vector space, $V^+ \subseteq V$ a pointed closed cone, and $\alpha_1, \ldots, \alpha_m \in V^*$ linear functionals. Define $C = \{x \in V^+ : \alpha_j(x) = b_j\}$. If x is an extreme point of C, then*

$$d_F = \dim\big(\mathrm{span}(\mathrm{face}_{V^+}(x))\big) \leq m.$$

Proof. By Corollary 11 (with $K = V^+$ viewed as a convex set and $\ell = m$), d_F is finite. Theorem 7 then gives $d_F \leq m$. □

6 Special Cases Recovered

1. **Classical Barvinok–Pataki** ($A = M_n(\mathbb{C})$). The unique irreducible representation is the identity, so $\pi_\varphi(A)' \cong M_r(\mathbb{C})$ where $r = \mathrm{rank}(\rho_\varphi)$. Hence $r^2 \leq m+1$, or $r^2 \leq m$ when one of the constraints is the trace normalisation already built into the spectrahedron.
2. **Carathéodory's theorem** ($A = C(X)$, abelian). All irreducible representations are one-dimensional, so the commutant decomposes as $\bigoplus_{j=1}^k \mathbb{C}$. The bound gives $k \leq m+1$: extreme measures under m moment constraints are supported on at most $m+1$ points.
3. **Block-diagonal algebras** ($A = \bigoplus_i M_{d_i}(\mathbb{C})$). The bound $\sum_i r_i^2 \leq m+1$ simultaneously constrains ranks within blocks *and* the number of active blocks.
4. **Infinite-dimensional** ($A = B(\mathcal{H})$). For normal states (density operators) under m constraints (including possibly unbounded Hamiltonians, handled via the Weis–Shirokov framework): $r^2 \leq m+1$. In particular, $m \leq 2$ constraints imply all extreme points are pure states.

7 Physical Example: Hydrogen Atom in a Magnetic Field

We illustrate the theorem with a concrete physical system: the hydrogen atom in an external magnetic field, subject to angular momentum constraints. This example demonstrates how the Barvinok–Pataki bound determines when pure-state variational methods suffice and when mixed states are unavoidable.

7.1 Hilbert space and quantum numbers

The non-relativistic hydrogen atom has Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ (spatial wave function tensored with spin-$\frac{1}{2}$). The Coulomb Hamiltonian

$$H_0 = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0\, r}$$

has eigenvalues $E_n = -13.6\ \mathrm{eV}/n^2$ for $n = 1, 2, 3, \ldots$, with simultaneous eigenstates $|n, l, m_l, m_s\rangle$ labelled by:

Quantum number	Range	Observable
n (principal)	$1, 2, 3, \ldots$	H_0 (energy)
l (orbital ang. mom.)	$0, 1, \ldots, n-1$	$\mathbf{L}^2 = l(l+1)\hbar^2$
m_l (magnetic)	$-l, -l+1, \ldots, l$	$L_z = m_l\hbar$
m_s (spin)	$\pm\frac{1}{2}$	$S_z = m_s\hbar$

The degeneracy of the n-th energy shell is $g_n = 2n^2$ (including spin). The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with $J_z = L_z + S_z$ having eigenvalue $m_j\hbar$ where $m_j = m_l + m_s$.

7.2 Zeeman Hamiltonian

In a uniform magnetic field $\mathbf{B} = B\hat{z}$, the interaction Hamiltonian is

$$H_Z = \frac{\mu_B B}{\hbar}\big(L_z + g_s\, S_z\big),$$

where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton and $g_s \approx 2$ is the electron spin g-factor. On the standard basis, $H_Z|n, l, m_l, m_s\rangle = \mu_B B(m_l + 2m_s)|n, l, m_l, m_s\rangle$. The full Hamiltonian is $H = H_0 + H_Z$.

7.3 The constrained optimisation problem

We seek the state of minimum energy subject to a constraint on the total angular momentum projection:

$$\min_{\rho}\ \mathrm{Tr}(H\rho) \qquad \text{subject to} \qquad \rho \geq 0,\ \ \mathrm{Tr}(\rho) = 1,\ \ \mathrm{Tr}(J_z\,\rho) = M\hbar. \tag{1}$$

This is a convex optimisation (linear objective over a convex constraint set). The minimum is attained at an extreme point of the feasible set

$$C = \big\{\rho \in \mathfrak{S}(\mathcal{H}) : \mathrm{Tr}(J_z\,\rho) = M\hbar\big\},$$

where $\mathfrak{S}(\mathcal{H})$ is the set of density operators on $\mathcal{H}$.

7.4 Applying the Barvinok–Pataki bound

Full Hilbert space ($A = B(\mathcal{H})$, infinite-dimensional). The constraint $\mathrm{Tr}(J_z\,\rho) = M\hbar$ constitutes $m = 1$ affine constraint on the state space $\mathfrak{S}(\mathcal{H})$. By Corollary 9 (or Corollary 13 applied to the positive cone in the trace-class operators),

$$r^2 \ \leq\ m + 1 = 2, \qquad \text{hence} \qquad r = 1.$$

The energy minimiser is a pure state.

More generally, with m constraints:

m	Constraints	Rank bound	Conclusion
1	$\mathrm{Tr}(J_z\,\rho) = M\hbar$	$r = 1$	Pure state
2	$+\ \mathrm{Tr}(\mathbf{J}^2\,\rho) = j(j+1)\hbar^2$	$r = 1$	Pure state
3	$+\ \mathrm{Tr}(\mathbf{L}^2\,\rho) = l(l+1)\hbar^2$	$r \leq 2$	Rank ≤ 2
4	$+\ \mathrm{Tr}(H_0\,\rho) = E$	$r \leq 2$	Rank ≤ 2

The transition from pure to potentially mixed occurs at $m = 3$: this is a direct consequence of the face dimension gap for $B(\mathcal{H})$, which has $\{1, 2\} \subseteq G(B(\mathcal{H}))$ (the face dimension spectrum is $\{0, 3, 8, 15, \ldots\}$).

Restriction to the $n = 2$ shell. For a concrete finite-dimensional calculation, restrict to the $n = 2$ shell. This has $g_2 = 2 \cdot 4 = 8$ states, so $\mathcal{H}_{n=2} \cong \mathbb{C}^8$ and $A = M_8(\mathbb{C})$:

l	m_l	m_s	$m_j = m_l + m_s$	$m_l + 2m_s$	**Subshell**
0	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2s$
0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2s$
1	-1	$+\frac{1}{2}$	$-\frac{1}{2}$	0	$2p$
1	-1	$-\frac{1}{2}$	$-\frac{3}{2}$	-2	$2p$
1	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2p$
1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2p$
1	1	$+\frac{1}{2}$	$+\frac{3}{2}$	2	$2p$
1	1	$-\frac{1}{2}$	$+\frac{1}{2}$	0	$2p$

The Coulomb Hamiltonian is $H_0 = E_2\, I_8$ (degenerate on the shell). Including spin–orbit coupling (fine structure), $H_{\text{fs}} = \xi_{21}\, \mathbf{L} \cdot \mathbf{S}$ splits the shell into multiplets: $2s_{1/2}$ ($l = 0$, $j = \frac{1}{2}$, 2 states), $2p_{1/2}$ ($l = 1$, $j = \frac{1}{2}$, 2 states), and $2p_{3/2}$ ($l = 1$, $j = \frac{3}{2}$, 4 states). The Zeeman term $H_Z = \mu_B B(m_l + 2m_s)$ is diagonal in the $|l, m_l, m_s\rangle$ basis.

The optimisation (1) restricted to this shell is an SDP on $M_8(\mathbb{C})$. Normalisation $\mathrm{Tr}(\rho) = 1$ is built into the state space. The Barvinok–Pataki bound gives:

m	**Constraints (on the $n = 2$ shell)**	$r^2 \le m + 1$	**Naïve bound**
1	$\mathrm{Tr}(J_z\, \rho) = M\hbar$	$r = 1$	$r \le 8$
2	$+\ \mathrm{Tr}(\mathbf{J}^2\, \rho) = j(j+1)\hbar^2$	$r = 1$	$r \le 8$
3	$+\ \mathrm{Tr}(\mathbf{L}^2\, \rho) = l(l+1)\hbar^2$	$r \le 2$	$r \le 8$
5	$+\ \mathrm{Tr}(S_z\, \rho) = s\hbar,\ \mathrm{Tr}(L_z\, \rho) = \mu\hbar$	$r \le 2$	$r \le 8$

7.5 Physical interpretation

1. **Automatic purity ($m \le 2$).** Fixing the total angular momentum projection M (and optionally j) is enough to guarantee that the energy minimiser is a pure state. This means pure-state variational methods (e.g. Rayleigh–Ritz, DMRG for the ground state) are *exact* for this problem — there is no advantage to working with density matrices.

2. **Low-rank mixed states ($m = 3$).** Adding a third constraint (e.g. fixing l) opens the door to rank-2 optimisers: the energy minimiser may be a *statistical mixture* of two pure states. But the rank is at most 2, not 8; thus only a two-dimensional search is needed, rather than optimisation over all 8×8 density matrices.

   This bound is **tight**: when the constraint values lie in the interior of the joint numerical range — i.e. values not achievable by any single pure state — the optimizer is genuinely mixed. For example, on the $n = 2$ shell with $m = 3$ constraints $\langle J_z \rangle = 0$, $\langle \mathbf{J}^2 \rangle = 2$, $\langle \mathbf{L}^2 \rangle = 1$ (values strictly between the eigenvalues of $\mathbf{J}^2$), the SDP optimizer has $\mathrm{rank}(\rho^*) = 2$ with eigenvalues $(\lambda_1, \lambda_2) \approx (0.569, 0.431)$. See Table 1 and Figure 1.

3. **Practical consequence.** For a numerical SDP solving (1), interior-point methods converge to the analytic centre (maximal rank), but the theorem guarantees that facial reduction or rank minimisation will find a solution of rank $\le \lfloor \sqrt{m+1} \rfloor$. This dramatically reduces the dimension of the search space: on the $n = 3$ shell ($g_3 = 18$, $\rho \in M_{18}(\mathbb{C})$), even $m = 10$ constraints yield $r \le 3$ rather than $r \le 18$.

Table 1: Genuine rank-2 optimizers on hydrogen shells with $m = 3$ constraints (BP bound $r \leq 2$). Constraint values are chosen in the interior of the joint numerical range, where no pure state is feasible.

n	dim	$\langle J_z \rangle$	$\langle \mathbf{J}^2 \rangle$	$\langle \mathbf{L}^2 \rangle$	rank	λ_1	λ_2
2	8	0	2.0	1.0	**2**	0.569	0.431
2	8	0	1.5	0.5	**2**	0.754	0.246
3	18	0	4.0	3.0	**2**	0.675	0.325

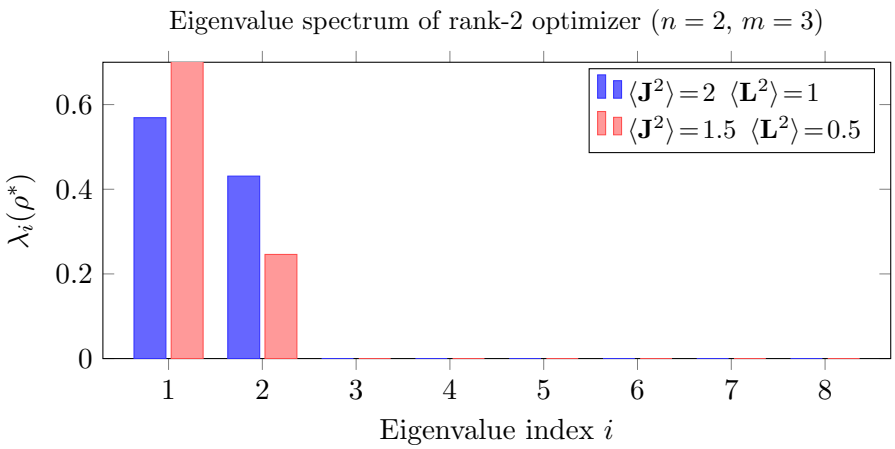

Figure 1: Eigenvalue spectra of two genuine rank-2 optimizers on the $n = 2$ shell (dim $= 8$) with $m = 3$ constraints and $\langle J_z \rangle = 0$. Exactly two eigenvalues are non-zero, saturating the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor = 2$. The constraint values lie in the interior of the joint numerical range and cannot be achieved by any pure state.

7.6 Numerical verification

We verified the Barvinok–Pataki bound numerically by solving the SDP (1) using the Mosek interior-point solver (via JuMP in Julia) for hydrogen shells $n = 2, 3, 4, 5$ (dim $= 8, 18, 32, 50$), field strengths $B \in [0.05, 5]$, and $m = 0, \ldots, 5$ angular momentum constraints. A two-phase approach was used: Phase 1 finds the optimal energy E^*; Phase 2 adds a small random Hermitian perturbation to the objective while pinning $\mathrm{Tr}(H\rho) = E^*$, generically selecting a unique extreme point of the optimal face.

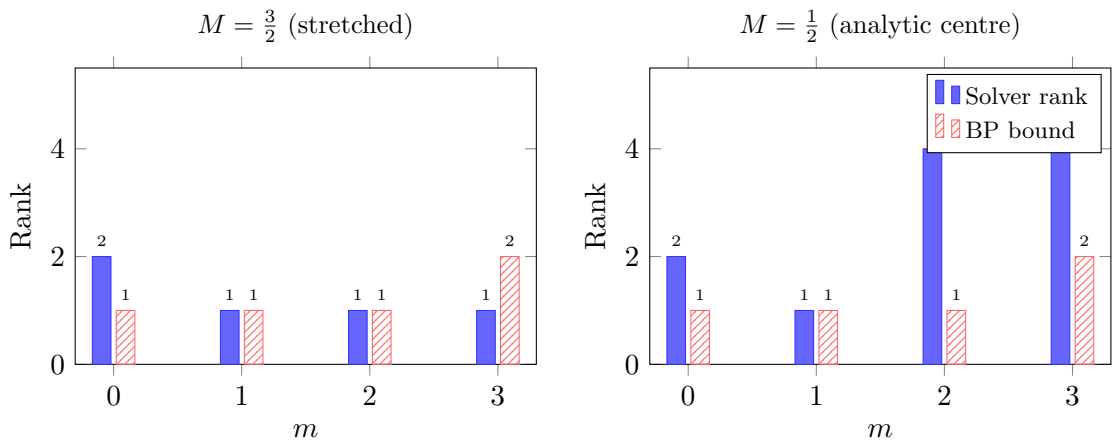

Figure 2: Optimizer rank on the $n = 2$ shell. **Left:** $M = \frac{3}{2}$ (maximally stretched); the extreme-point optimizer satisfies the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor$. **Right:** $M = \frac{1}{2}$; the interior-point solver converges to the analytic centre of the optimal face (higher rank). Facial reduction would recover an extreme point satisfying BP.

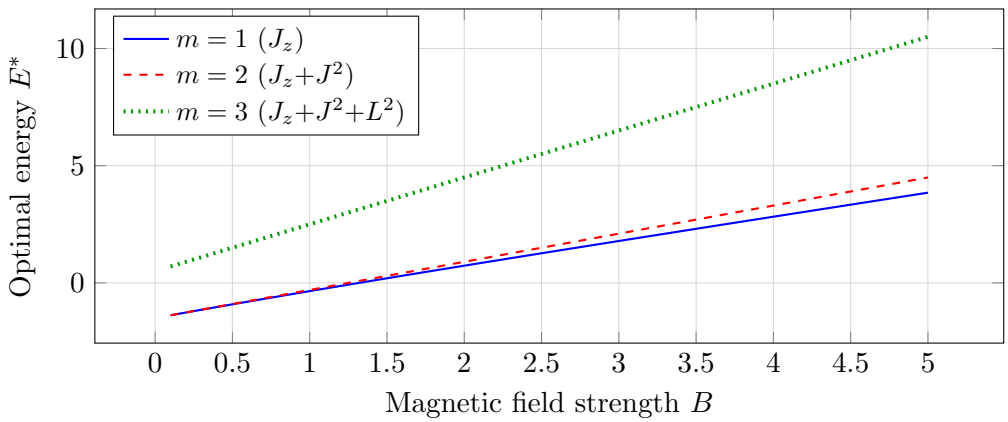

Figure 3: Optimal energy vs. field strength on the $n = 3$ shell (dim $= 18$, $M = \frac{3}{2}$). Adding constraints raises the minimum energy. All optimizers are pure ($r = 1$), consistent with BP.

8 Cone-Valued Inequality Constraints

The results above treat scalar affine equality constraints $\alpha_j(x) = b_j$. A natural extension, proposed during the follow-up discussion, is to allow *conic inequality constraints* where the constraint maps take values in ordered vector spaces.

Theorem 14 (Barvinok–Pataki for conic inequalities)**.** *Let (V, V^+) be an ordered vector space with pointed cone, and let I be a finite index set. For each $i \in I$, let (W_i, W_i^+) be an ordered*

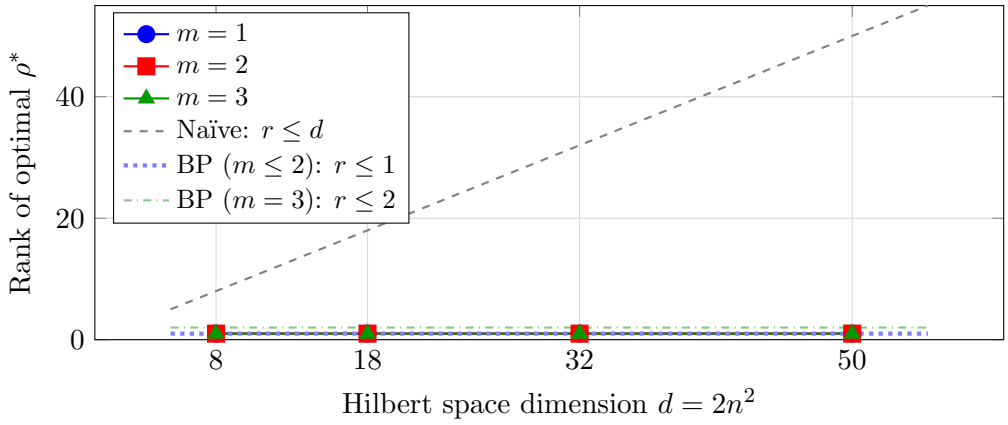

Figure 4: Optimizer rank vs. Hilbert space dimension ($B = 1$, $M = \frac{3}{2}$). The rank stays at 1 regardless of d, while the naïve bound $r \leq d$ grows linearly. For $n = 5$ ($d = 50$), BP reduces the rank bound from 50 to at most 2.

vector space and $\Phi_i \colon V \to W_i$ *a linear map. For given* $a_i \in W_i$*, define the feasible set*

$$C = \left\{ x \in V^+ \;\middle|\; \Phi_i(x) \geq_{W_i^+} a_i \;\; \forall\, i \in I \right\},$$

where $\Phi_i(x) \geq_{W_i^+} a_i$ *means* $\Phi_i(x) - a_i \in W_i^+$*. Set* $m = \sum_{i \in I} \dim(W_i)$*. If* x *is an extreme point of* C *and* $d_F = \dim(\operatorname{span}(\operatorname{face}_{V^+}(x)))$ *is finite, then*

$$\boxed{d_F \;\leq\; m.}$$

Proof. Define the combined constraint map $\Phi \colon V \to \bigoplus_i W_i$ by $\Phi(v) = (\Phi_i(v))_{i \in I}$. Let $F = \operatorname{face}_{V^+}(x)$ and consider the restriction $\Phi_F \colon \operatorname{span}(F) \to \bigoplus_i W_i$.

By rank–nullity, $d_F = \dim(\ker \Phi_F) + \dim(\operatorname{im} \Phi_F) \leq \dim(\ker \Phi_F) + m$.

It suffices to show $\ker \Phi_F = \{0\}$ when x is extreme in C. Suppose $D \in \ker \Phi_F$, $D \neq 0$. Then $\Phi_i(D) = 0$ for all i, so

$$\Phi_i(x + tD) - a_i = (\Phi_i(x) - a_i) + t\,\Phi_i(D) = \Phi_i(x) - a_i \in W_i^+$$

for all $t \in \mathbb{R}$. Hence $x + tD$ satisfies all conic inequality constraints whenever $x + tD \in V^+$. The remainder of the argument (perturbation along D, exit from V^+ by pointedness, contradiction with extremality) follows identically to Steps 2–4 of Theorem 7. □

Remark 15 (Equality vs. inequality constraints)**.**

1. **Equalities.** If the constraints are equalities $\Phi_i(x) = a_i$, each can be decomposed into $\dim(W_i)$ scalar equalities and the bound $d_F \leq m$ reduces to Theorem 7.

2. **Simplex cones ($W_i^+ \cong \mathbb{R}_+^{k_i}$).** A conic inequality $\Phi_i(x) \geq_{\mathbb{R}_+^{k_i}} a_i$ is equivalent to k_i scalar inequality constraints. The conic bound $d_F \leq \sum k_i$ coincides with the scalar bound.

3. **Non-simplex cones (e.g. $W_i^+ = \mathrm{PSD}_{k_i}$).** The conic bound $d_F \leq \sum \dim(\mathrm{Sym}_{k_i})$ cannot be recovered by converting each inequality to scalar constraints, because the PSD cone is not generated by a simplex.

4. **Inactive constraints.** If a conic constraint is *inactive* at x (i.e. $\Phi_i(x) - a_i$ lies in the interior of W_i^+), then any perturbation D preserves this constraint for small $|t|$, regardless of $\Phi_i(D)$. Thus inactive constraints do not contribute to the kernel condition. More precisely, if k constraints are inactive, then $d_F \leq m'$ where $m' = \sum_{i:\,\text{active}} \dim(W_i)$.

Remark 16 (Slack variable formulation)**.** Alternatively, one can introduce slack variables $S_i = \Phi_i(x) - a_i \in W_i^+$ and rewrite the conic inequality as an equality $\Phi_i(x) - S_i = a_i$ over the product cone $V^+ \times \prod_i W_i^+$. The face of the product cone at $(x, (S_i)_i)$ has dimension $d_F(x) + \sum_i d_F(S_i \text{ in } W_i^+)$, and the equality formulation gives

$$d_F(x) + \sum_i d_F(S_i) \;\le\; \sum_i \dim(W_i).$$

This is at least as tight as Theorem 14 whenever some slack S_i has a non-trivial face (i.e. the constraint is not fully active). However, the slack formulation requires tracking the face structure of each S_i, which may not be accessible a priori. Moreover, practical face-reduction algorithms (iterating the perturbation argument) cannot control whether they reduce $d_F(x)$ or $d_F(S_i)$ at each step.

9 Application to Spectrahedra

A *spectrahedron* is the feasible set of a semidefinite program, typically defined by one or more linear matrix inequalities (LMIs). Theorem 14 applies directly.

Corollary 17 (Block-diagonal LMI spectrahedra)**.** *Let $x \in \mathbb{R}^n$ and consider two LMI constraints*

$$F(x) = F_0 + \textstyle\sum_{i=1}^n x_i F_i \succeq 0, \qquad G(x) = G_0 + \textstyle\sum_{i=1}^n x_i G_i \succeq 0,$$

with $F_i \in \mathrm{Sym}_k$, $G_i \in \mathrm{Sym}_p$. If x^* is an extreme point of the spectrahedron $S = \{x : F(x) \succeq 0,\, G(x) \succeq 0\}$, then

$$\frac{(r_F + r_G)(r_F + r_G + 1)}{2} \;\le\; n,$$

where $r_F = \mathrm{rank}(F(x^*))$ and $r_G = \mathrm{rank}(G(x^*))$. This is strictly tighter than the individual bounds $r_F(r_F+1)/2 \le n$ and $r_G(r_G+1)/2 \le n$.

Proof. The combined LMI $\mathrm{diag}(F(x), G(x)) \succeq 0$ is a single $(k+p) \times (k+p)$ LMI parameterised by n variables. The standard Barvinok–Pataki bound gives the result. The tightening over individual bounds follows from $(r_F + r_G)^2 > r_F^2 + r_G^2$ when both ranks are positive. □

Example 18 (Entanglement witness rank bound)**.** Let $d = d_A \cdot d_B$ and consider the cone of PPT (positive partial transpose) matrices:

$$\mathcal{P} = \big\{\, W \in \mathrm{PSD}_d \;\big|\; W^{\Gamma_B} \succeq 0 \,\big\},$$

where Γ_B denotes partial transpose on subsystem B. This is a pointed closed cone (if $W, -W \in \mathcal{P}$, then $W = 0$). The constraint $W^{\Gamma_B} \succeq 0$ is a conic inequality with $\Phi = \Gamma_B$, $W_1^+ = \mathrm{PSD}_d$, $\dim(W_1) = d(d+1)/2$ (real) or d^2 (complex Hermitian).

Consider optimising over $\mathcal{P}$ subject to m scalar affine constraints (e.g. normalisation $\mathrm{Tr}(W) = 1$ and $\mathrm{Tr}(A_j W) = b_j$). By Theorem 14, any extreme point W^* satisfies

$$\mathrm{rank}(W^*)^2 \;\le\; m \quad \text{and} \quad \mathrm{rank}(W^{*\Gamma_B})^2 \;\le\; m,$$

where the first bound comes from the face of PSD_d and the second from the face of the PPT cone.

For $m = 1$ (normalisation only): $\mathrm{rank}(W^*) = 1$ and $\mathrm{rank}(W^{*\Gamma_B}) = 1$. The extreme PPT witnesses under normalisation alone are rank-1 with rank-1 partial transpose, i.e. they are product projectors $|\psi\rangle\langle\psi| \otimes |\phi\rangle\langle\phi|$.

Numerical verification (see `numerics/spectrahedra_bp.jl`) for a 2×2 system confirms: $m = 1$ gives $\mathrm{rank}(W) = \mathrm{rank}(W^\Gamma) = 2$; $m = 8$ gives $\mathrm{rank}(W) = 4$, $\mathrm{rank}(W^\Gamma) = 3$.

10 Inequality Constraints and the Role of Slack

The interplay between equality and inequality constraints merits further discussion, following observations by Lennart Binkowski and Timo Ziegler.

Example 19 (PSD matrix with scalar inequalities)**.** Consider $X \in \mathrm{PSD}_n$ (real symmetric) with m scalar inequality constraints $\mathrm{Tr}(A_i X) \leq b_i$. Introducing slack variables $s_i \geq 0$ with $\mathrm{Tr}(A_i X) + s_i = b_i$, this becomes a problem over the product cone $\mathrm{PSD}_n \times \mathbb{R}^m_+$ with m equality constraints.

The face of the product cone at (X, s) has dimension

$$d_F(X, s) = \frac{r(r+1)}{2} + |\{i : s_i > 0\}|,$$

where $r = \mathrm{rank}(X)$. The Barvinok–Pataki bound gives

$$\frac{r(r+1)}{2} + |\{i : s_i > 0\}| \;\leq\; m. \tag{2}$$

In particular, if all constraints are active ($s_i = 0$ for all i), then $r(r+1)/2 \leq m$ (the equality-constraint bound). If k constraints are inactive, then $r(r+1)/2 \leq m-k$, confirming the heuristic that inactive constraints "free up" dimensions.

However, a face-reduction algorithm iterating the perturbation argument of Theorem 7 cannot guarantee which term in (2) decreases at each step: it may reduce $\mathrm{rank}(X)$ or merely set some s_i to zero.

11 Obstructions to Nonlinear Extensions

It is natural to ask whether the Barvinok–Pataki bound extends to nonlinear constraints. Two perspectives emerged from the discussion.

Remark 20 (Manifold framework)**.** If the equality constraints involve smooth nonlinear functions $f_i \colon V \to \mathbb{R}$ and the target values $(b_1, \ldots, b_m)$ are regular values, then by the implicit function theorem the constraint set $C = \{x \in V^+ : f_i(x) = b_i\}$ is a smooth manifold of codimension m. The perturbation direction D lives in the kernel of the differential $Df|_x$ restricted to the tangent cone of V^+ at x. The rank–nullity argument applies to the differential, giving a *local* dimension bound.

However, it is not clear whether the resulting flow $t \mapsto x + tD + O(t^2)$ reaches a face of V^+ of strictly lower dimension, which is the analogue of Step 3 in the proof of Theorem 7.

Remark 21 (Obstruction to localisation)**.** The Barvinok–Pataki argument fundamentally relies on *linearity* of the constraints: the perturbation $x + tD$ preserves the constraint *exactly* (not merely to first order), so the entire line segment from x to the boundary of V^+ is feasible.

For nonlinear constraints, this fails. A nonlinear constraint can force a feasible point to lie deep in the interior of a high-dimensional face, with no feasible path to the boundary. For example, consider a polyhedron P with a nonlinear constraint that confines x to a "shell" strictly inside P: every point satisfying the constraint is interior to P, so no perturbation reaches a proper face.

Linear constraints avoid this because they generate faces upon intersection with the cone: the hyperplane $\{x : \alpha(x) = b\}$ slices the cone into proper faces.

Acknowledgements

The proof idea underlying this work is due to Timo Ziegler, who presented the classical Barvinok–Pataki argument for PSD matrices during a blackboard discussion at the Institut für Theoretische Physik, Leibniz Universität Hannover, on 3 March 2026. His formulation of the perturbation–extremality argument was subsequently recognised to generalise directly to pointed cones and Archimedean order unit spaces. Martin Plavala recognised during the discussion that the argument extends to general cones. Alexander Stottmeister pointed out that the face-dimension quantity d_F can be controlled in infinite dimensions via Theorem 3.3 of Weis–Shirokov [4], which requires only a real vector space and a convex set, without operator-algebraic assumptions. The recording was transcribed and typeset with the assistance of Claude (Anthropic).

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